



Santiago Ignacio Romero Tufiño (26 years old)



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Electromechanical Engineer

Electrical Design | Industrial Automation | Mechanical CAD Design

Electromechanical Engineer specialized in electrical design, industrial automation and CAD modeling. Experience in control and power engineering, Siemens and Schneider PLC/HMI programming, electrical panel design, development of plans for infrastructure projects and mechanical modeling using AutoCAD, AutoCAD electrical and Inventor.

WORK EXPERIENCE

2024-2026	Electrical, Control & Automation Project Engineer Electrocapital
• Development of electrical engineering for control and power systems, including electrical diagrams, single-line diagrams, design of electrical panels, and supervision of assembly and installation. • Design of electrical installations for public and private projects, including buildings of the Departmental Prosecutor's Office of Chuquisaca and Santa Cruz, using AutoCAD and AutoCAD Electrical. • Programming of Siemens PLCs (S7-1200 and LOGO! 8.3) and Schneider Electric (M221 and M241), integrating HMI interfaces for process supervision and control. • Design and implementation of drive systems using Siemens SIRIUS 3RW50/3RW55 soft starters and variable frequency drives (VFDs) applied to industrial processes and wastewater treatment plants (WWTPs).	
2025	Electromechanical Specialist "Accidental Association A & T Associates"
• Development of the electrical design for the Technical Study of Pre-investment Design of the Bicentennial Plaza of Bolivia (Cobija). Demand calculation, sizing of the electrical system and preparation of lighting plans, outlets, power, grounding, electrical panels and single-line diagrams. Using CAD and Revit software.	
2023 – 2026	Mechanical Designer "Freelance"
• Parametric design and modeling of mechanical parts, mechanisms, and assemblies using Autodesk Inventor. Development of test stations for automation with PLCs, including structures, mechanical supports, conveyor systems and pneumatic mechanisms. Design of automated systems for food classification and customized mechanisms aimed at manufacturing and automation. • Preparation of manufacturing drawings, technical documentation and 3D models for production processes.	
2025-2026	3D Design & Printing & Robotics Instructor "U.S. Embassy American Spaces"
• I designed and taught courses in 3D Design, programming, robotics and home automation for maker space in American spaces, training students in parametric modeling with Autodesk Inventor, prototyping and 3D printing, in addition to the development of automation projects with Arduino, sensors and actuators.	
2022- 2023	Electrical Design Engineering "Rofer, Engineering and Services." "Esdesur, Services and Business" (Labor Practices)
• I designed and implemented low voltage electrical projects and industrial facilities such as INTI laboratory warehouses and redesign of the ENTEL building, applying technical standards and efficiency criteria, which reduced installation times and optimized energy distribution in productive environments. using CAD software.	

EDUCATION

2018-2024	Electromechanical Engineering Academic Diploma in National Provision (2024) • Obtaining 1st place in the ranking of graduates of the careers of the Faculty of Engineering and Applied Mechanical-Electronic Sciences of the promotion of January/2024. • Degree project approved in mention of congratulations from the teaching tribunal 92/100 pts. (2023) Universidad Mayor Real Y Pontificia De San Francisco Xavier De Chuquisaca, Sucre- Bolivia	
2025	Postgraduate Course - Renewable Energies for the Energy Transition in South America: Technical, Economic and Geopolitical Aspects • Certificate of Completion of the Postgraduate Continuing Education Course (100 academic hours) Universidad Mayor Real Y Pontificia De San Francisco Xavier De Chuquisaca, Sucre- Bolivia	
2025	Aspire Leaders Program – Founded by Harvard University • I completed the academic program developing skills awarded by the Aspire Institute-Harvard	
English as a Foreign Language Bolivian American Center CBA - Advanced (C1)		Spanish (Native)

ABILITIES

Electrical and mechanical design	AutoCAD	Management of CAD software focused on general design and drawings for electrical, electronic, mechanical design projects.
	AutoCAD Electrical	Management of CAD software focused on the design of electrical control planes, automation and protection, as well as 2d electrical panels
	Inventor	Management of CAD software focused on the general design of parts, components, mechanisms or machines.
Automation Control (PIC-HMI)	Aunt Portal	S7-1200 PLC programming concepts and skills, HMI.
	Logo Softcomfort	PLC programming concepts and skills with the Siemens logo.
	Simaris Project-Design	Selection and sizing of low-voltage electrical components for panel design.
	EcoStruxure	M221-M241 PLC Programming Concepts and Skills
	Vijeo Designer	Monitoring System Design with Vijeo Designer-HMI
	Ecodial	Pre-engineering design in electrical projects.
Lighting	DiaLux Evo 13	Management of software focused on the simulation and calculation of lighting for industrial and residential spaces and environments.
Office	Microsoft Office	Office management in general: Excel, Word, among others.

CERTIFICATES

2024	Industrial Automation Course. Electrocapital Engineering training
2024	AutoCAD Electrical Course - Electrical Design, Drawings for Automation AutoCAD Electrical – Drawing Design for Automation and Electrical Systems and AutoCAD Electrical. Training Ideas Engineering training
2024	Multi-brand PLC programming courses (Logo!, S7-1200 - Siemens, M221 - Schneider) Training of industrial automation processes: from basic and advanced programming of PLCs (Logo!, S7-1200 and M221). Management of variable frequency drives, instrumentation and control fundamentals, the use of Ecodial for electrical design and the integration of solutions through M241 PLC. Together, it forms a complete journey for designing, programming, and monitoring modern automated systems. Training Ideas Engineering training
2025	Course on selection and sizing of low voltage electrical components for panel design. Simaris Certificate Program 2025 Siemens
2024	Electrical Installations Course with Revit. Construction Academy Insight
2018	National Congress of Electrical Engineering and Electronic Engineering Royal and Pontifical University of San Francisco Xavier de Chuquisaca, Sucre- Bolivia

OUTSTANDING ACHIEVEMENTS

2025	Military High School "Tte. Edmundo Andrade" Evaluation Jury and Collaborator in Defense of Robotics Projects and Institutional Support.
2018-2024	Faculty of Engineering and Applied Sciences Mecca-Electronics - Winner of the Teaching Assistant Scholarship for 4 consecutive years (2020-2023) - Obtaining 1st place in the ranking of graduates of the careers of the Faculty of Engineering and Applied Mechanical-Electronic Sciences of the promotion of January/2024.
2020-2021	Student Scientific Society Faculty of Mechanical-Electrical-Electronic Engineering Founding member of the Student Scientific Society of the Faculty of Mechanical-Electrical-Electronic Engineering of the Royal and Pontifical University of San Francisco Xavier de Chuquisaca.
2025	Winner of the Inspira National Alumni Project Round-Meeting Granted by the U.S. Embassy in Bolivia.
2016-2022	Access Scholarship Winner and Access Plus Scholarship Winner Awarded by the U.S. Embassy in Bolivia and the Ciudad Mundo Foundation.